

# Meeting brief — Priority PII detection model

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## Results

The 1,000-trial search produced a fast, recall-first document tagger that met the critical safety requirement on the frozen evaluation matrix. The selected model reads the first 1,000 characters and passed **55 of 55 measurable priority tag–corpus recall gates conclusively**. Its worst observed priority recall was **98.88%**, and its lowest 95% bootstrap lower bound was **98.11%**.

The model is fast: **2.20 ms p95 per document on one CPU core**, or **919 documents/second** on the fixed 1,000-document benchmark sample. It is 4.01× faster at p95 than the same fusion model reading 20,000 characters.

The important trade-off is precision. Equal-corpus macro F2 is **0.4835**, well below the 0.90 aspiration, and micro F1 is **0.3812**. The recommended decision is therefore to approve this model for **internal high-recall triage or shadow evaluation**, with downstream verification, but not for autonomous redaction, blocking, compliance certification, or registry-backed production deployment.

| Executive measure       | Outcome                 | Decision impact                                             |
|-------------------------|-------------------------|-------------------------------------------------------------|
| Priority recall gates   | 55/55 conclusive passes | Critical recall objective met on measurable holdout pairs   |
| Worst priority recall   | 98.88%                  | Above the requested 90% floor                               |
| Lowest recall 95% bound | 98.11%                  | Recall result is statistically conclusive on measured pairs |
| Equal-corpus macro F2   | 0.4835                  | Precision/overall quality target not met                    |
| Micro F1 tie-break      | 0.3812                  | Reported, but did not override the recall-first decision    |
| One-core p95 latency    | 2.20 ms/document        | Passes the fast operating-point requirement                 |
| One-core throughput     | 919.1 documents/second  | Suitable for high-volume first-pass triage                  |
| Package reproduction    | Exact over 126,129 rows | Shipped loader matches the sealed model                     |

## TL;DR

- Exactly 1,000 MLflow-tracked trials were completed across hash, TF-IDF, EmbeddingBag, and fusion candidates.
- The selected 1k fusion exceeded 90% recall for every measurable priority tag–corpus pair, including SSN, ITIN, medical identifiers, financial account identifiers, identity documents, credentials, names, and addresses.
- The 1k pass is the best speed/recall operating point: 4.01× faster than 20k while preserving all 55 point gates.
- Macro F2 remains 0.4835. Approve only for internal triage/shadow use with downstream review; do not present it as a complete redaction system.

- Before production, resolve long-document coverage, precision, source-data licensing, registry promotion, and live monitoring.

## Meeting metadata

| Field              | Value                                                                                     |
|--------------------|-------------------------------------------------------------------------------------------|
| Date               | 2026-08-25                                                                                |
| Suggested duration | 30 minutes                                                                                |
| Meeting objective  | Decide the allowed use of the 1k PII model and fund the next validation/improvement phase |
| Model              | pii-priority-fusion-1k-v1                                                                 |
| Scope              | 61 document-level PII/PCI/PHI tags; 16 recall-priority tags                               |
| Evaluation         | Eight holdout corpora scored separately; 126,129 indexed rows                             |
| Status             | Packaged research/internal candidate; not registry-promoted or deployed                   |
| Source commit      | d662a3a                                                                                   |

## Decisions requested

- 1 **Approve internal use?** Approve the 1k model for high-recall triage and shadow evaluation only, with downstream verification and audit logging.
- 2 **Long-document policy?** Require upstream chunking or a longer secondary scan whenever sensitive content can occur after character 1,000.
- 3 **Quality target?** Keep macro F2 0.90 as a future target, or define a realistic operating threshold and per-tag precision floors for the next cycle.
- 4 **Licensing owner?** Assign review of AI4Privacy and NonCommercial source restrictions before redistribution or commercial use.
- 5 **Production path?** Decide whether to fund registry promotion, service integration, live traffic validation, and monitoring after the above items pass.

## Speed and read-depth discussion

All latency numbers below are end-to-end, one-core measurements on the same deterministic 1,000-document sample. Quality was independently recalculated on all eight holdouts at each read depth.

| Read pass | p95 latency | Docs/s | Speed vs 20k | Macro F2 | Priority point gates | Worst recall |
|-----------|-------------|--------|--------------|----------|----------------------|--------------|
| 1,000     | 2.200 ms    | 919.1  | 4.01×        | 0.4835   | 55/55                | 0.9888       |

| Read pass | p95 latency | Docs/s | Speed vs 20k | Macro F2 | Priority point gates | Worst recall |
|-----------|-------------|--------|--------------|----------|----------------------|--------------|
| 2,500     | 4.211 ms    | 570.1  | 2.10×        | 0.4797   | 55/55                | 0.9686       |
| 10,000    | 6.766 ms    | 409.9  | 1.30×        | 0.4799   | 55/55                | 0.9966       |
| 20,000    | 8.829 ms    | 354.4  | 1.00×        | 0.4798   | 55/55                | 0.9966       |

The 1k pass saves 6.63 ms/document at p95 relative to 20k and slightly improves macro F2 by 0.0036. It reduces worst observed priority recall by 0.0079, but the final 1,000-resample bootstrap still placed every measurable priority recall lower bound above 90%.

The speed result must not be interpreted as late-document coverage. The selected artifact intentionally ignores content after character 1,000. Upstream chunking is the recommended protection when long documents are in scope.

## Priority recall coverage

The 16 requested priority categories all had measurable evidence on at least one holdout corpus, although the number of measurable corpora differs by tag.

- Government and identity: SSN, ITIN, passport, driver's licence, military ID, and visa number.
- Medical: MRN, health-plan beneficiary number, and patient ID.
- Financial: bank account number, credit-card number, and IBAN.
- Credentials: password and PIN.
- Person and location: full name and address.

Across these categories, the minimum per-tag recall ranged from 98.88% to 100% on measurable corpora. Address was the weakest measured category at 98.88%; its lowest 95% bound was 98.11%. Patient ID was measurable in only one corpus, so its apparent 100% recall has narrower external evidence than name or address.

## What worked

- Recall-first selection prevented higher-F2 candidates from winning when they missed critical tags. The standalone hash-F2, TF-IDF, and EmbeddingBag models had stronger micro F1 but failed between five and six priority gates.
- Locking priority labels to the recall-max head preserved the safety gate while per-label fusion improved non-priority quality.
- Exact overlap controls excluded 22,816 training rows sharing evaluation text; evaluation data were never removed or relabelled.
- Separate corpus scoring prevented a strong corpus from hiding weak results.
- The model remained compact and CPU-only, with a portable NumPy runtime.

## What did not meet target

- Equal-corpus macro F2 reached 0.4835, not 0.90. False positives remain the main quality problem.
- BetterDataAI was the weakest complete holdout at macro F2 0.2313. OpenPII was the strongest at 0.6297 and should not be quoted as general performance.
- Full 10k and 20k fusion passes exceeded the 5 ms p95 latency aspiration at 6.77 ms and 8.83 ms respectively.
- The model emits document tags, not value locations. It cannot perform redaction or explain which text produced a prediction.

## Risks and safeguards

| Risk                        | Current safeguard                                   | Required next step                                                        |
|-----------------------------|-----------------------------------------------------|---------------------------------------------------------------------------|
| False positives             | Recall-first model used only as a first-pass signal | Add downstream verifier/span detector and per-tag precision floors        |
| PII after character 1,000   | Limitation stated in model card                     | Chunk documents or run a longer secondary pass                            |
| Dataset shift and OCR noise | Eight heterogeneous holdouts                        | Validate on representative live documents and OCR slices                  |
| Partial/silver labels       | Missing labels never treated as negatives           | Add human-reviewed complete-catalogue evaluation data                     |
| Licensing restrictions      | Bundle marked research/internal only                | Complete legal/data-rights review before redistribution or commercial use |
| Automated misuse            | No deployment or registry promotion performed       | Require human review and documented use-case approval                     |

## Recommended decision

Approve pii-priority-fusion-1k-v1 as an **internal research and shadow-mode champion candidate** for document-level, high-recall triage. Do not promote it to a production registry alias or use it for autonomous redaction until the following exit conditions are satisfied:

- A representative live-data evaluation confirms the priority recall gate.
- A chunking/secondary-scan design covers long documents.
- Per-tag precision/F2 requirements are agreed and met.
- Source-data usage and redistribution rights are cleared.
- Monitoring, rollback, audit logging, and human-review workflows are defined.

## Suggested 30-minute agenda

| Time      | Topic                                       | Desired outcome                                                 |
|-----------|---------------------------------------------|-----------------------------------------------------------------|
| 0–5 min   | Goal and recommendation                     | Align on internal-only recall-first positioning                 |
| 5–10 min  | Recall evidence                             | Confirm that the 55/55 gate result is sufficient for shadow use |
| 10–15 min | Speed/read-depth trade-off                  | Approve 1k default plus long-document fallback                  |
| 15–22 min | Precision, licensing, and operational risks | Agree on blocking issues for production                         |
| 22–27 min | Decisions requested                         | Record approvals, rejections, and constraints                   |
| 27–30 min | Owners and dates                            | Assign next actions                                             |

## Decision and action log

Complete this table during the meeting.

| Item                                | Decision / action | Owner | Due date | Status   |
|-------------------------------------|-------------------|-------|----------|----------|
| Internal shadow/triage use          |                   |       |          | Open     |
| Long-document chunking policy       |                   |       |          | Open     |
| Precision/F2 acceptance criteria    |                   |       |          | Open     |
| Live representative evaluation set  |                   |       |          | Open     |
| Licensing and redistribution review |                   |       |          | Open     |
| Registry/deployment decision        |                   |       |          | Deferred |
| Monitoring and rollback design      |                   |       |          | Deferred |

## Verification

- The package re-predicted all 126,129 indexed holdout rows through its own `tagger.py` entry point.
- Packaged and sealed predictions were byte-identical with SHA-256  
3bfd3f776d8b7c09a7f11daafb8b5c4d2316571e724232506857be2f0168c194.
- Equal-corpus macro F2 reproduced exactly at 0.48345749382010117.
- Thirty-three focused tests and lint checks passed.
- ZIP integrity and all bundle member checksums passed.
- Final bundle SHA-256: cd8462d346f9f3b9c095a08ffe2c7711a9ea49ec2e677d123df8bb72cd0fd192.

## Artifacts

- Full technical report: 26-08-25\_priority-recall-1000-run.md.

- Experiment workbook: 26-08-25\_Experiment-Log.xlsx.
- Complete command archive: 26-08-25\_priority-recall-1000-run.commands.txt.
- Read-depth evidence: ../benchmarks/read\_depth.json.
- Package verification: ../dist/bundle\_verification.json.
- Delivery bundle: ../dist/pii-priority-fusion-1k-v1.zip.